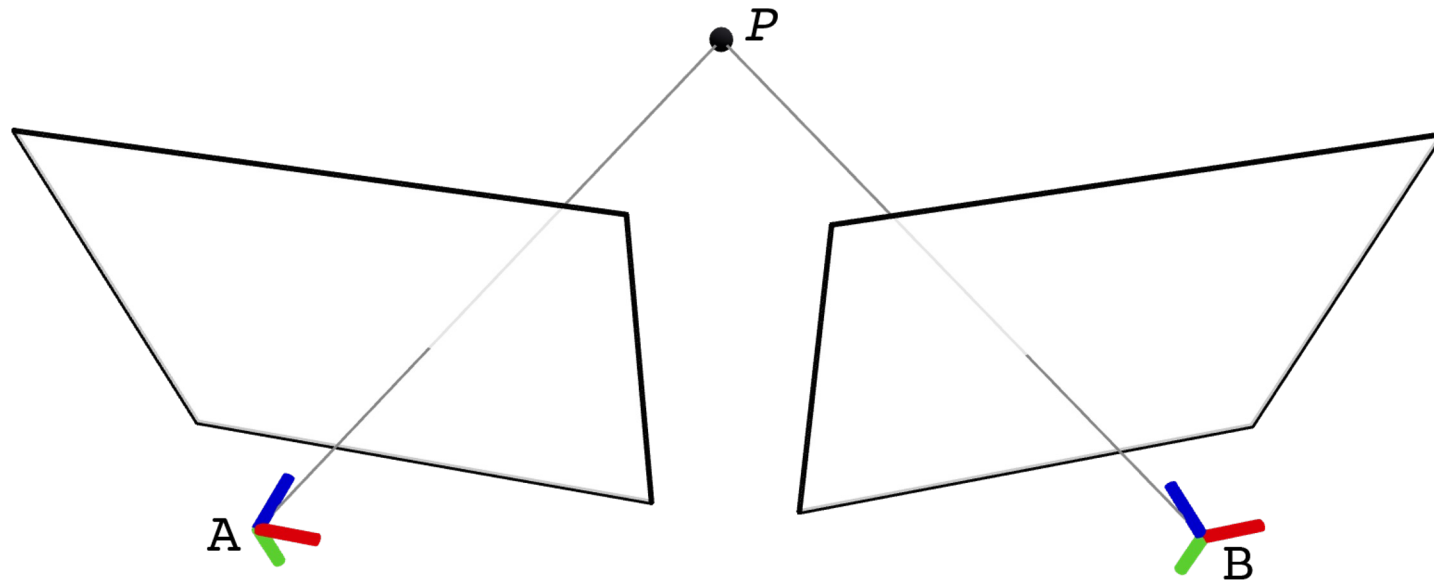


CS 180 Discussion #8

World, Cameras, Pixels, Rays



Welcome!!

Logistics

- Midterm grades this week
- Project 4 released

Agenda

3D coordinate systems

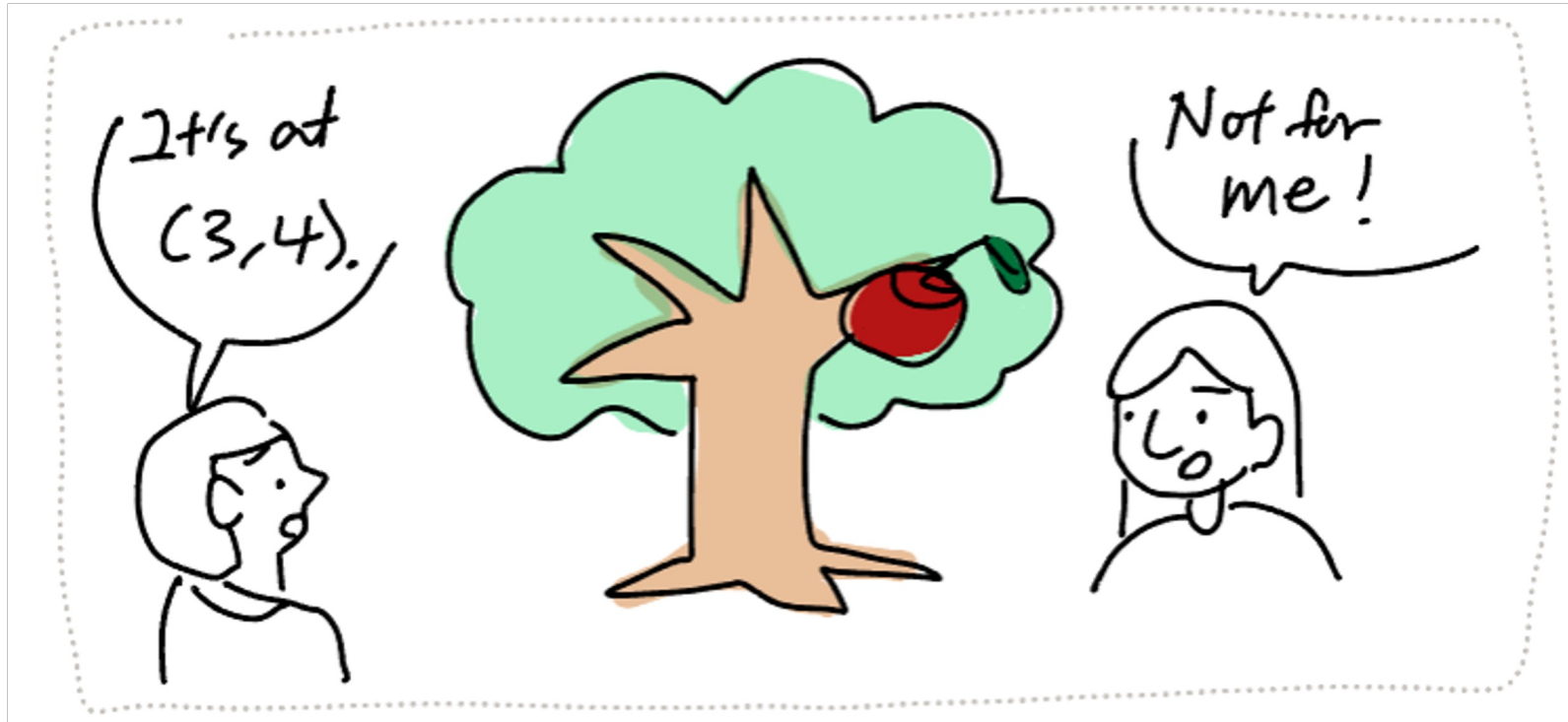
World, Camera, Pixel

Many cameras

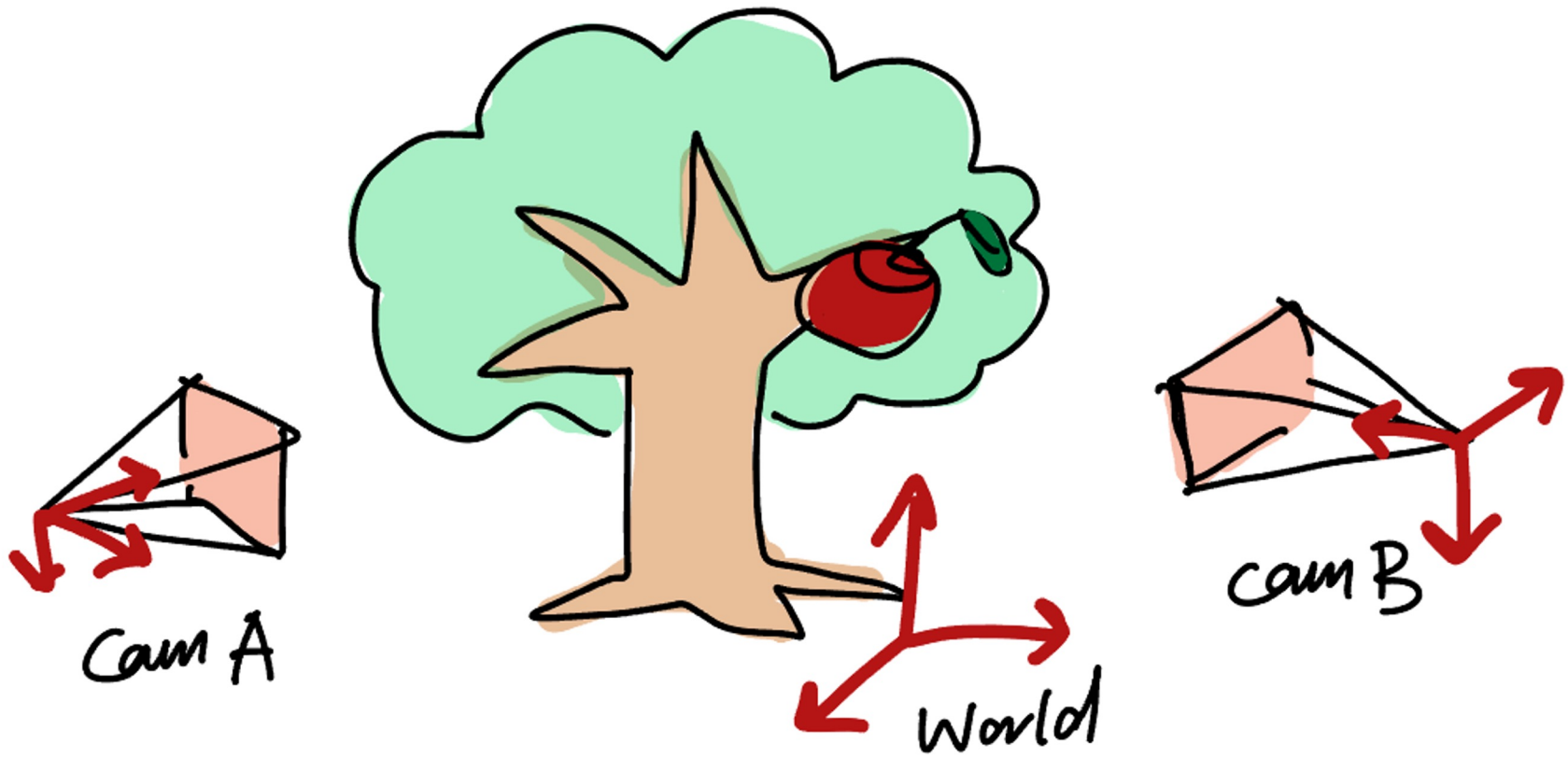
Rays

Coordinate frames matter!

How many times do you say “**your right hand side, or my side?**”

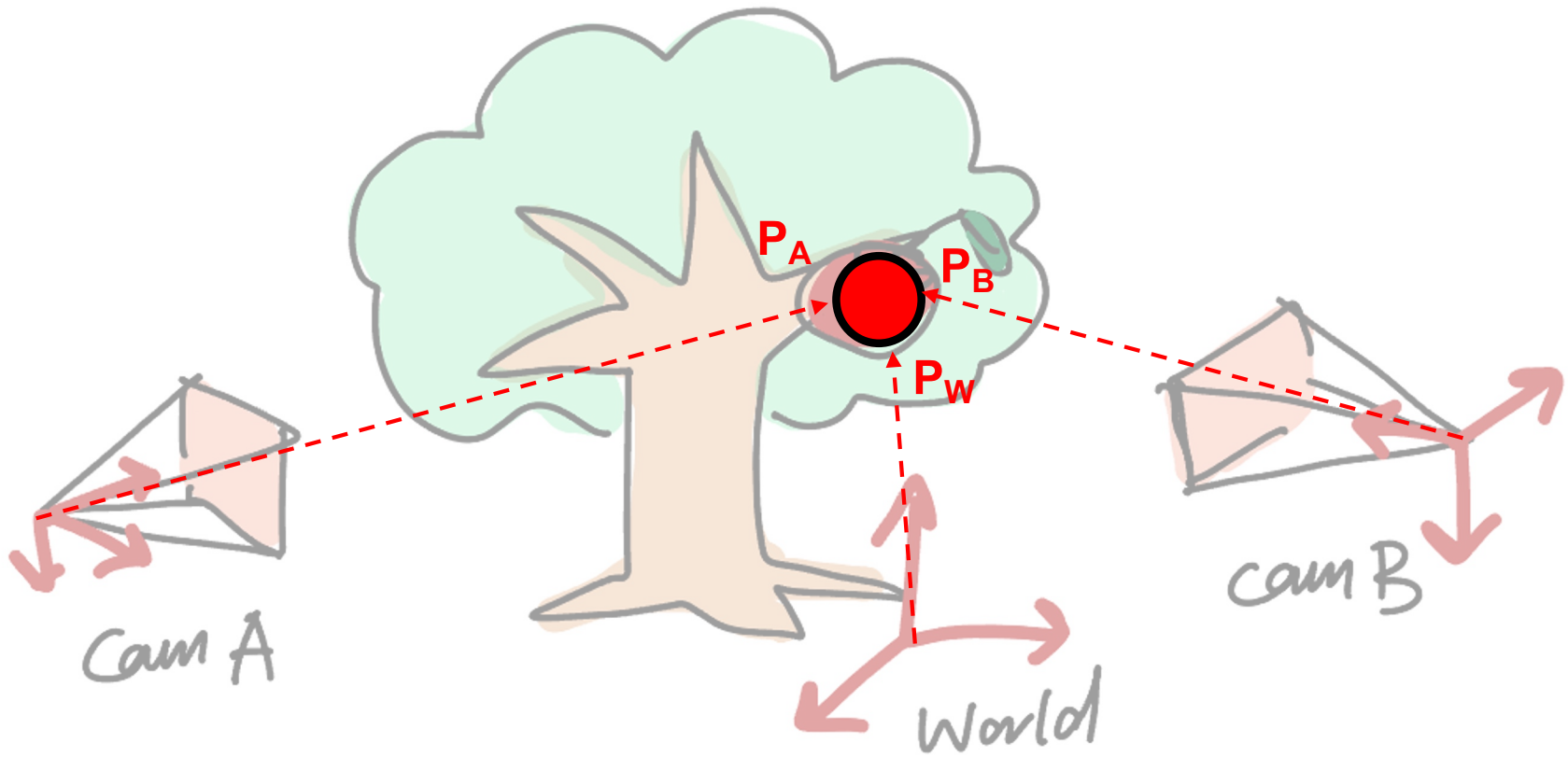


Frames (each is a frame of reference)



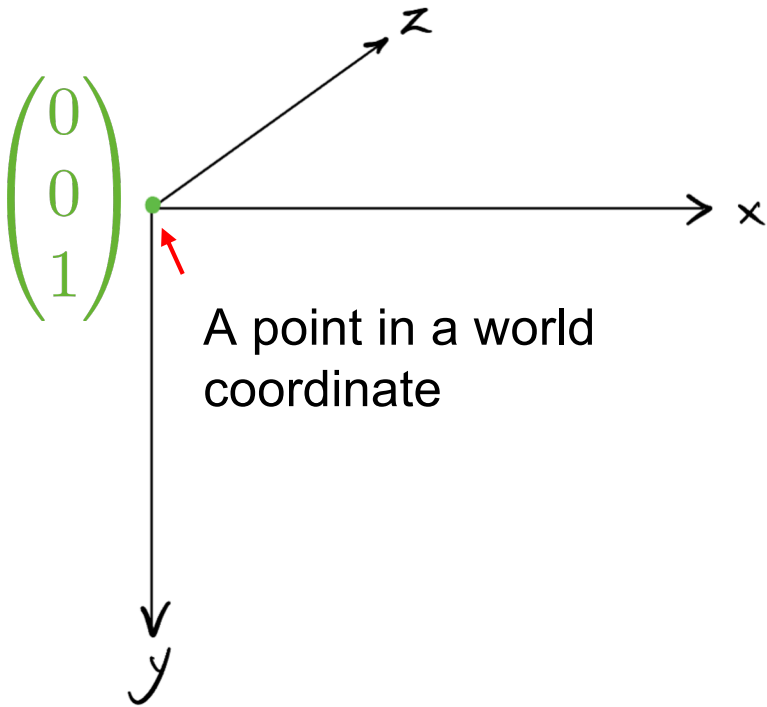
*World = another camera

Same point can be observed from different frames

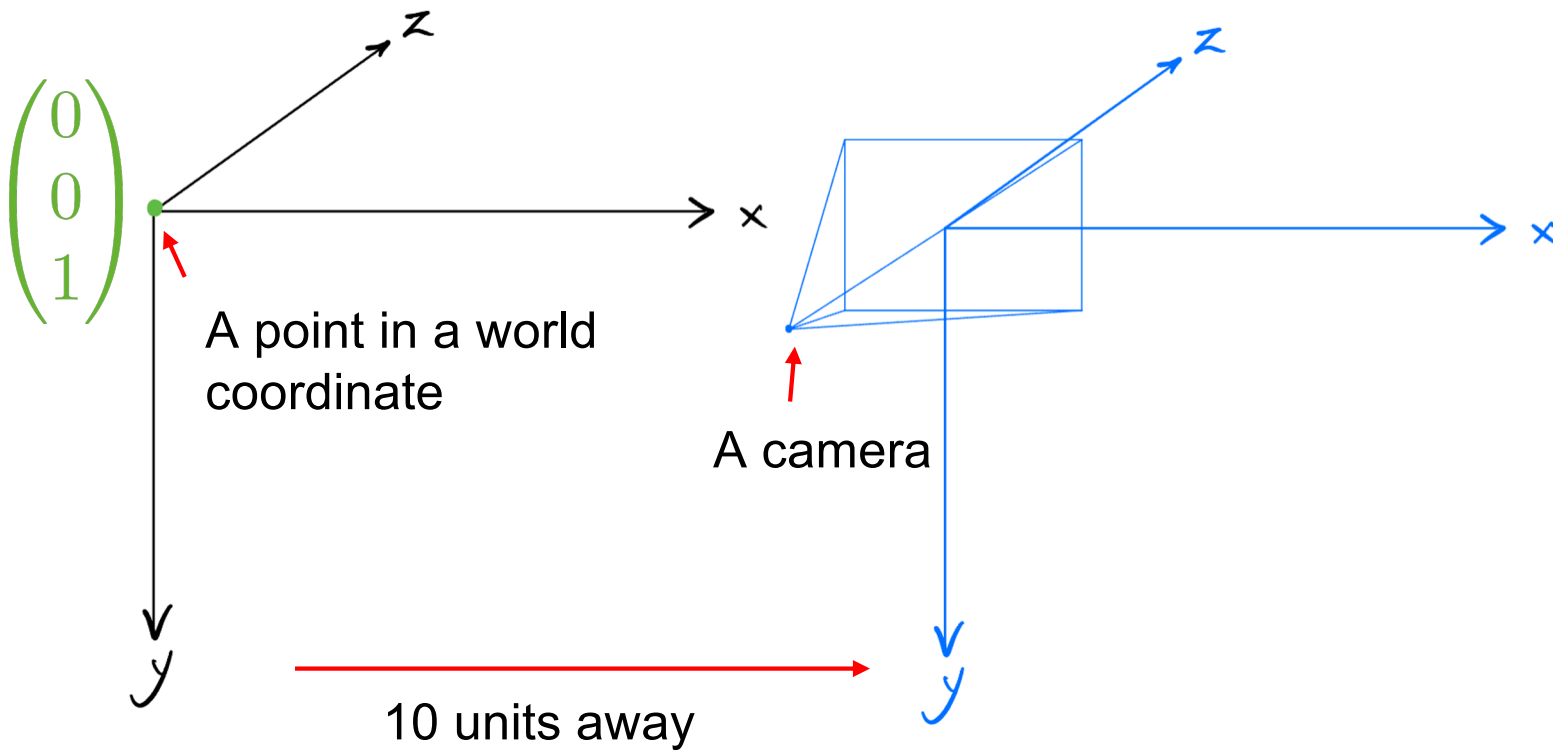


Examples

World



Camera

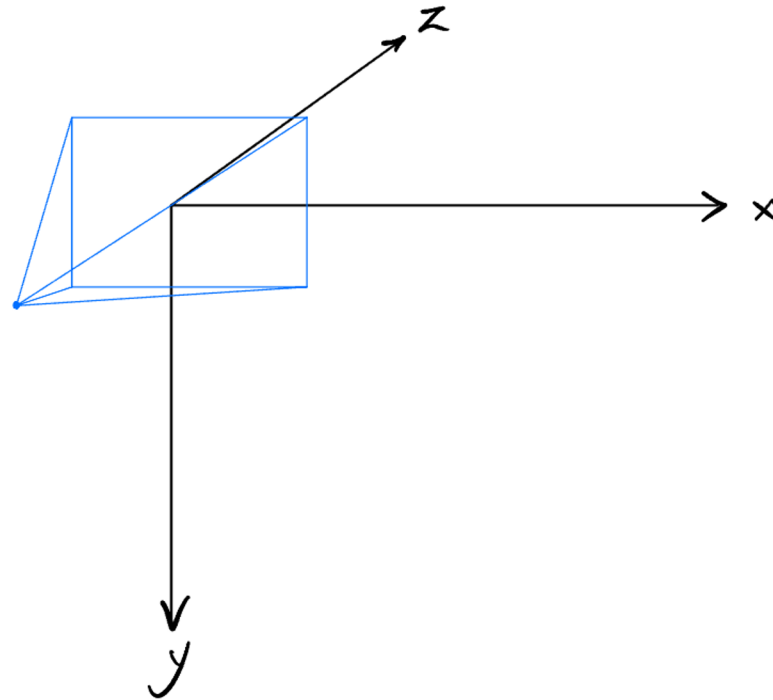


*Where is the point in "camera coordinate"?

World => Camera coordinate

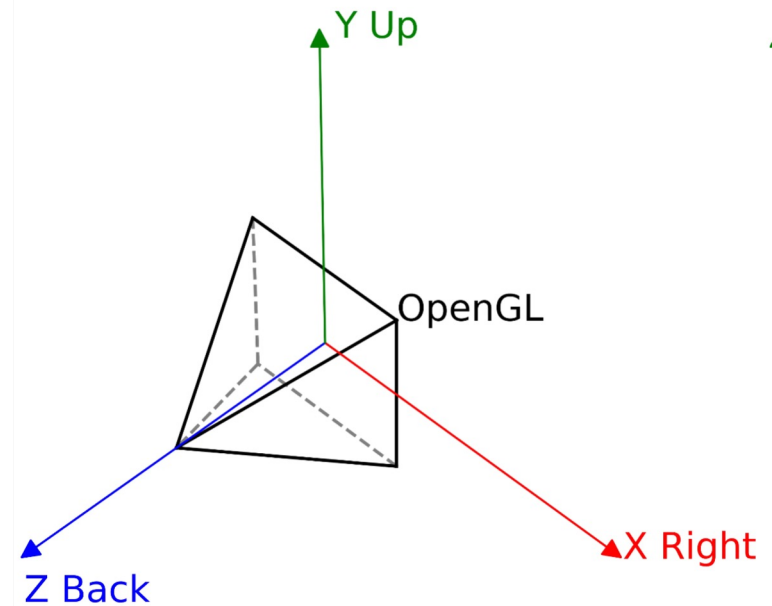
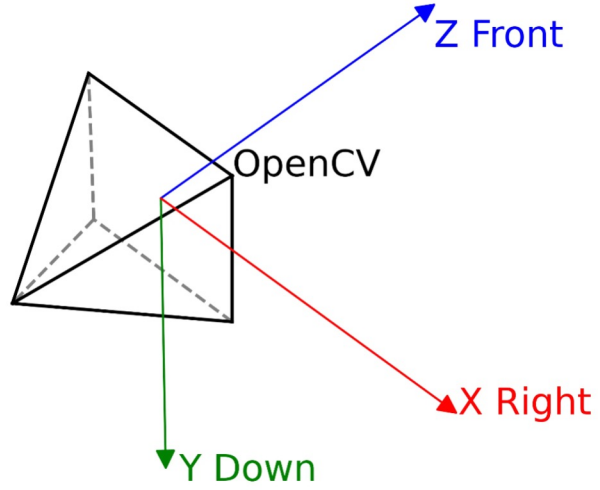
$$\begin{pmatrix} -10 \\ 0 \\ 1 \end{pmatrix}$$

Same point in the
camera coordinate
(shifted -10 units)



*Good luck imaging rotation 😊

Camera conventions



*We use OpenCV, but there are more...

World to camera transformation

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \longrightarrow \begin{pmatrix} -10 \\ 0 \\ 1 \end{pmatrix}$$

World Camera

World to camera transformation

World to camera transformation

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} -10 \\ 0 \\ 1 \end{pmatrix}$$

World Camera

$$\underbrace{\begin{pmatrix} -10 \\ 0 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{x}_c} = \underbrace{(\quad)}_{T_{w \rightarrow c}} \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{x}_w}$$


Homogeneous

$$\begin{pmatrix} -10 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & -10 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}_{4 \times 4} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

World to camera transformation

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \longrightarrow \begin{pmatrix} -10 \\ 0 \\ 1 \end{pmatrix}$$

World Camera

$$\underbrace{\begin{pmatrix} -10 \\ 0 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{x}_c} = \underbrace{(\quad)}_{T_{w \rightarrow c}} \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{x}_w}$$

$$\begin{pmatrix} -10 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \boxed{\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}} & \boxed{\begin{matrix} -10 \\ 0 \\ 0 \end{matrix}} \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

Rotation Translation

World to camera transformation

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \longrightarrow \begin{pmatrix} -10 \\ 0 \\ 1 \end{pmatrix}$$

World Camera

$$\underbrace{\begin{pmatrix} -10 \\ 0 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{x}_c} = \underbrace{(\quad)}_{T_{w \rightarrow c}} \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{x}_w}$$

$$\begin{pmatrix} -10 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \boxed{\mathbf{R}} & \boxed{\mathbf{t}} \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

World to camera transformation

$$\begin{array}{ccc}
 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} & \longrightarrow & \begin{pmatrix} -10 \\ 0 \\ 1 \end{pmatrix} \\
 \text{World} & & \text{Camera}
 \end{array}
 \qquad
 \underbrace{\begin{pmatrix} -10 \\ 0 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{x}_c} = \underbrace{(\quad)}_{T_{w \rightarrow c}} \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{x}_w}$$

Shorthand

$$\mathbf{x}_c = \mathbf{T}_{w \rightarrow c} \mathbf{x}_w$$

$$\begin{pmatrix} x_c \\ 1 \end{pmatrix} = \begin{pmatrix} \boxed{\mathbf{R}} & \boxed{\mathbf{t}} \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_w \\ 1 \end{pmatrix}$$

*Lenient usage of homogeneous coordinates

World to camera transformation

$$\mathbf{x}_c = \mathbf{T}_{w \rightarrow c} \mathbf{x}_w$$

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} -10 \\ 0 \\ 1 \end{pmatrix}$$

World

Camera

$$\mathbf{T}_{w \rightarrow c} = \begin{pmatrix} 1 & 0 & 0 & -10 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Camera to world

$$\mathbf{x}_w = \mathbf{T}_{w \rightarrow c} \mathbf{x}_c$$

$$\begin{pmatrix} -10 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Camera

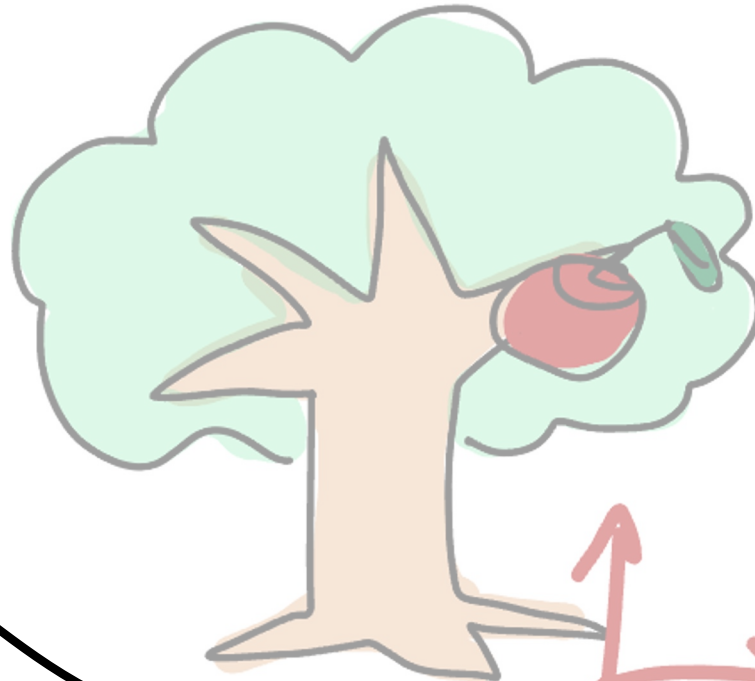
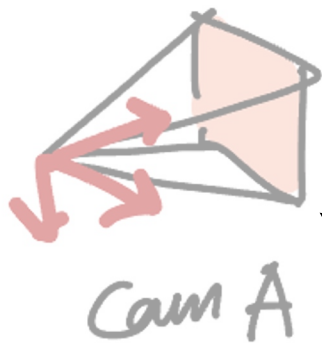
World

$$\mathbf{T}_{c \rightarrow w} = \mathbf{T}_{w \rightarrow c}^{-1} = \begin{pmatrix} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

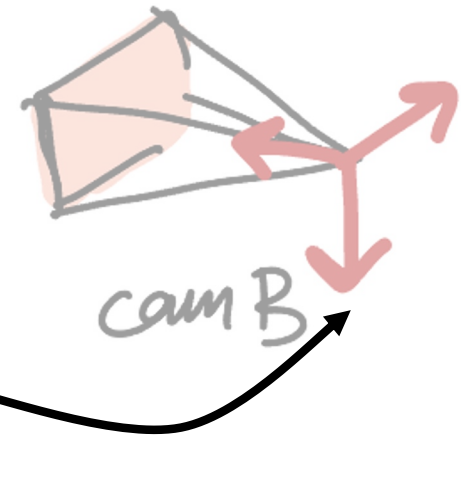
Also called “camera extrinsics”

Transformations take you **from** one frame **to** another

T_{WA} (or $T_{W \leftarrow A}$)
from: A, to: world

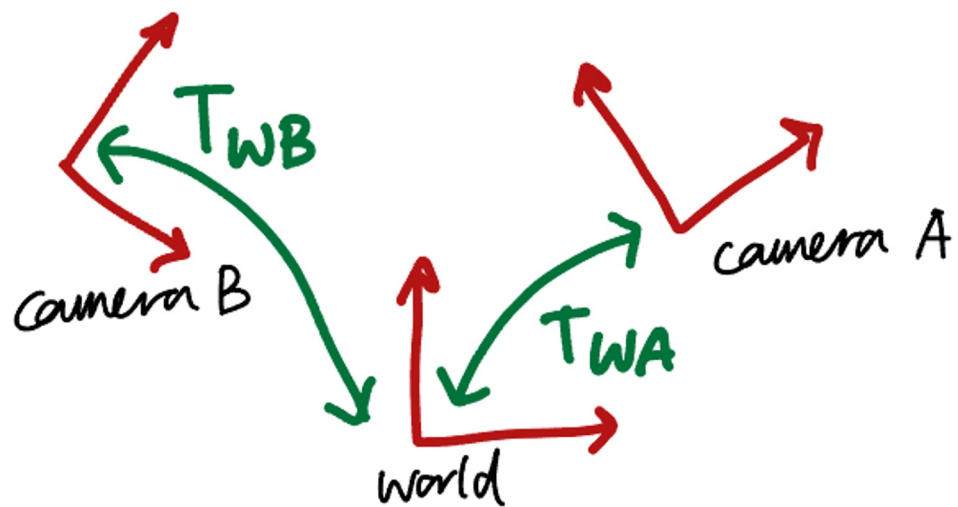


T_{WB} (or $T_{W \leftarrow B}$)
from: B, to: world

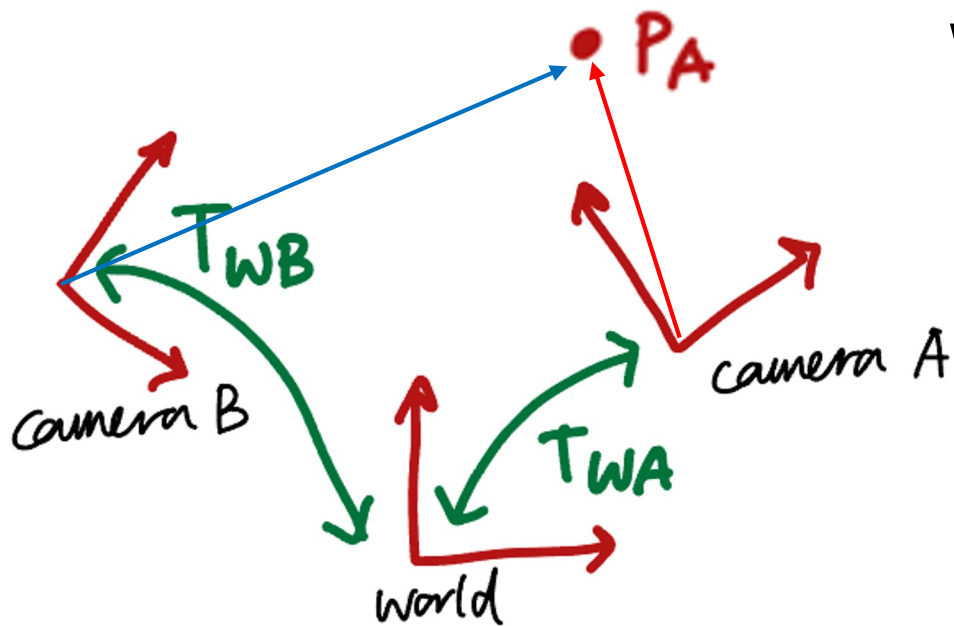


*It's a convention, don't blame us!

Chaining transformations



Chaining transformations



What is P_B ?

$$\begin{aligned} P_B &= T_{BW} T_{WA} P_A \\ &= \boxed{T_{WB}^{-1}} T_{WA} P_A \end{aligned}$$

Inverse!

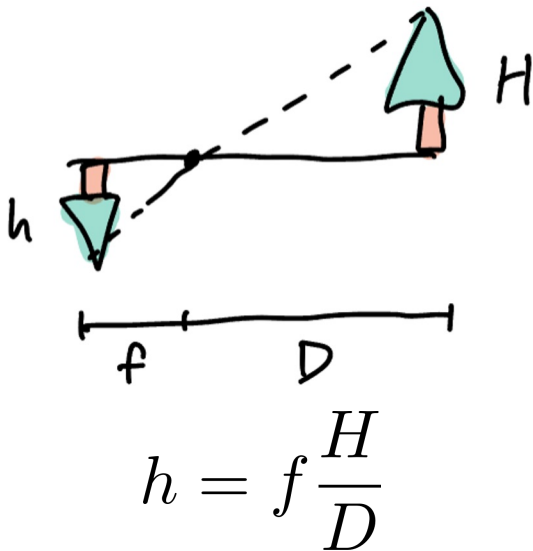
Do part 1a-d

Will continue on pinhole camera in 3D

Pinhole cameras + 3D

Our pinhole camera equations transform 3D point to 2D

They assume point is in camera frame



$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

Sensor offset

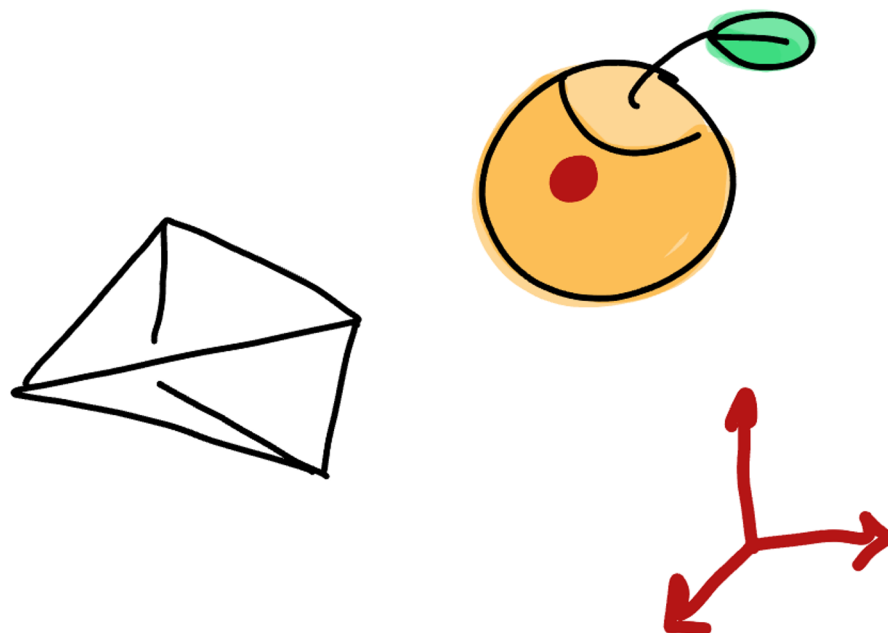
$$\lambda \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = K \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Pixel coordinate

Point in 3D

Example exercise!

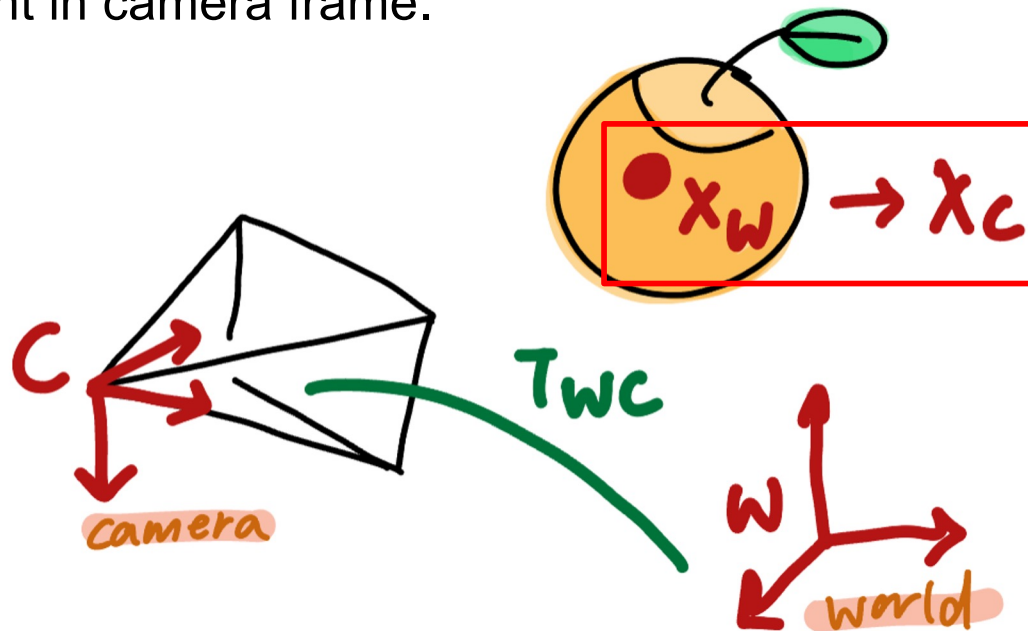
Let's put a point on this orange onto an image.



Example exercise!

Let's put a point on this orange onto an image.

1. Put the point in camera frame.



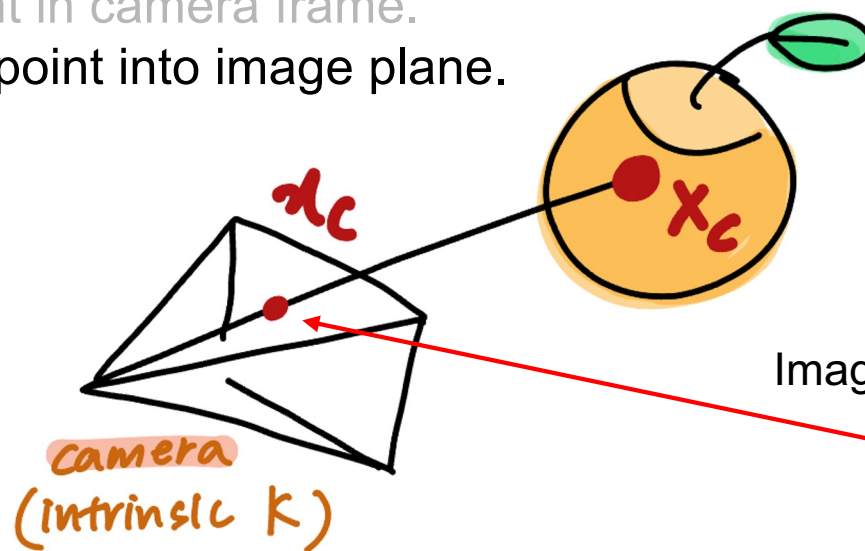
Point in camera frame

$$\begin{aligned}\mathbf{X}_c &= \mathbf{T}_{cw} \mathbf{X}_w \\ &= (\mathbf{T}_{wc})^{-1} \mathbf{X}_w\end{aligned}$$

Example exercise!

Let's put a point on this orange onto an image.

1. Put the point in camera frame.
2. Project the point into image plane.



Point in camera frame

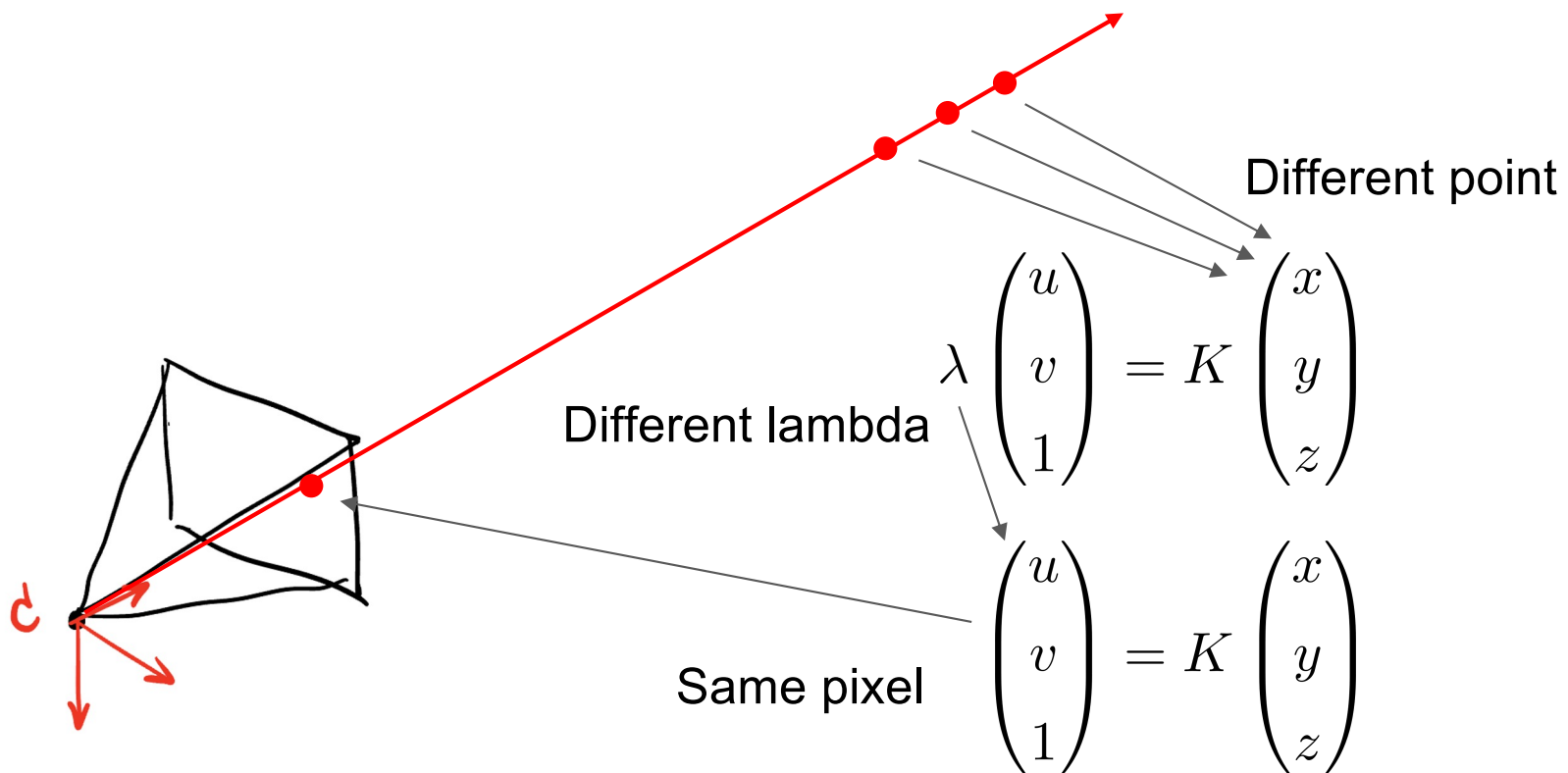
$$\begin{aligned}\mathbf{X}_c &= \mathbf{T}_{cw} \mathbf{X}_w \\ &= (\mathbf{T}_{wc})^{-1} \mathbf{X}_w\end{aligned}$$

$$\lambda \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = K \mathbf{X}_c$$

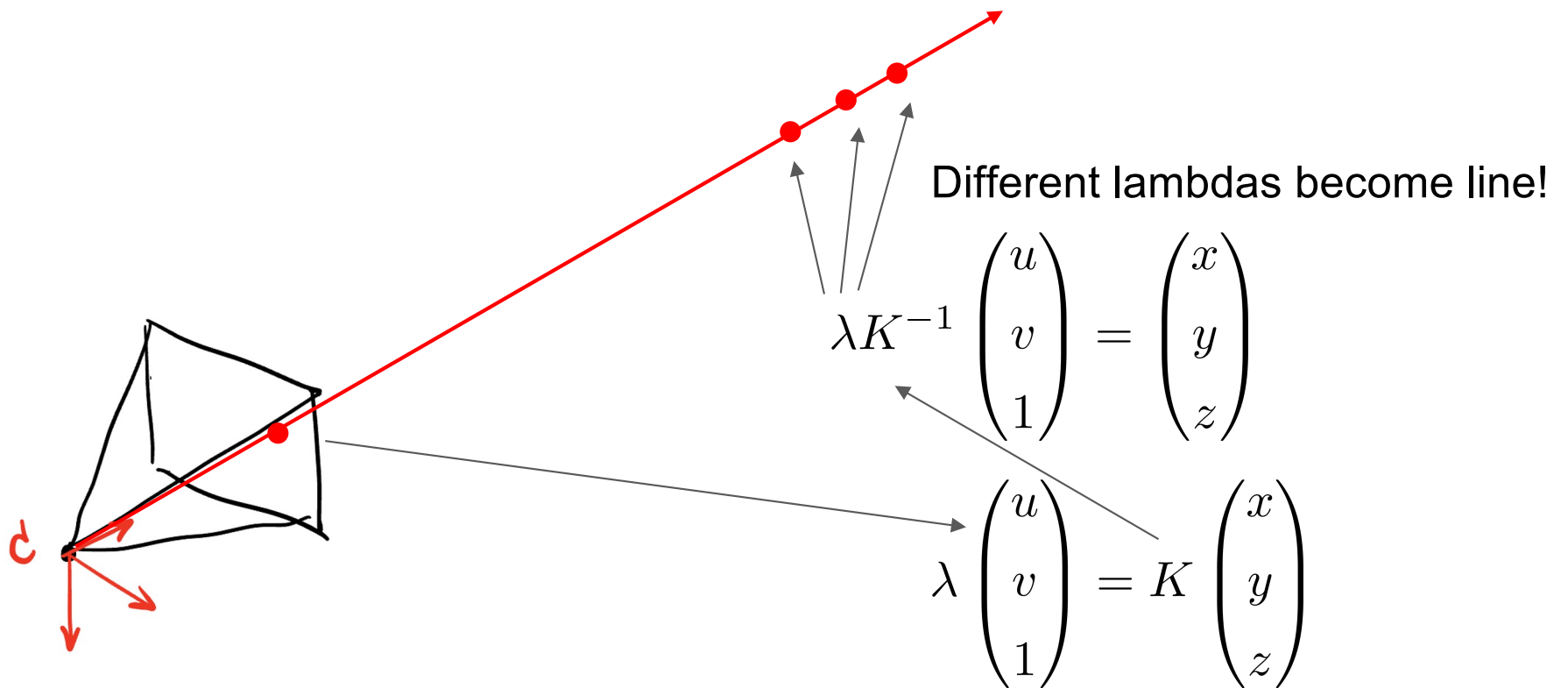
A red arrow points from the $\mathbf{X}_c$ term in the equation above to the $\mathbf{X}_c$ term in this equation.

Rays

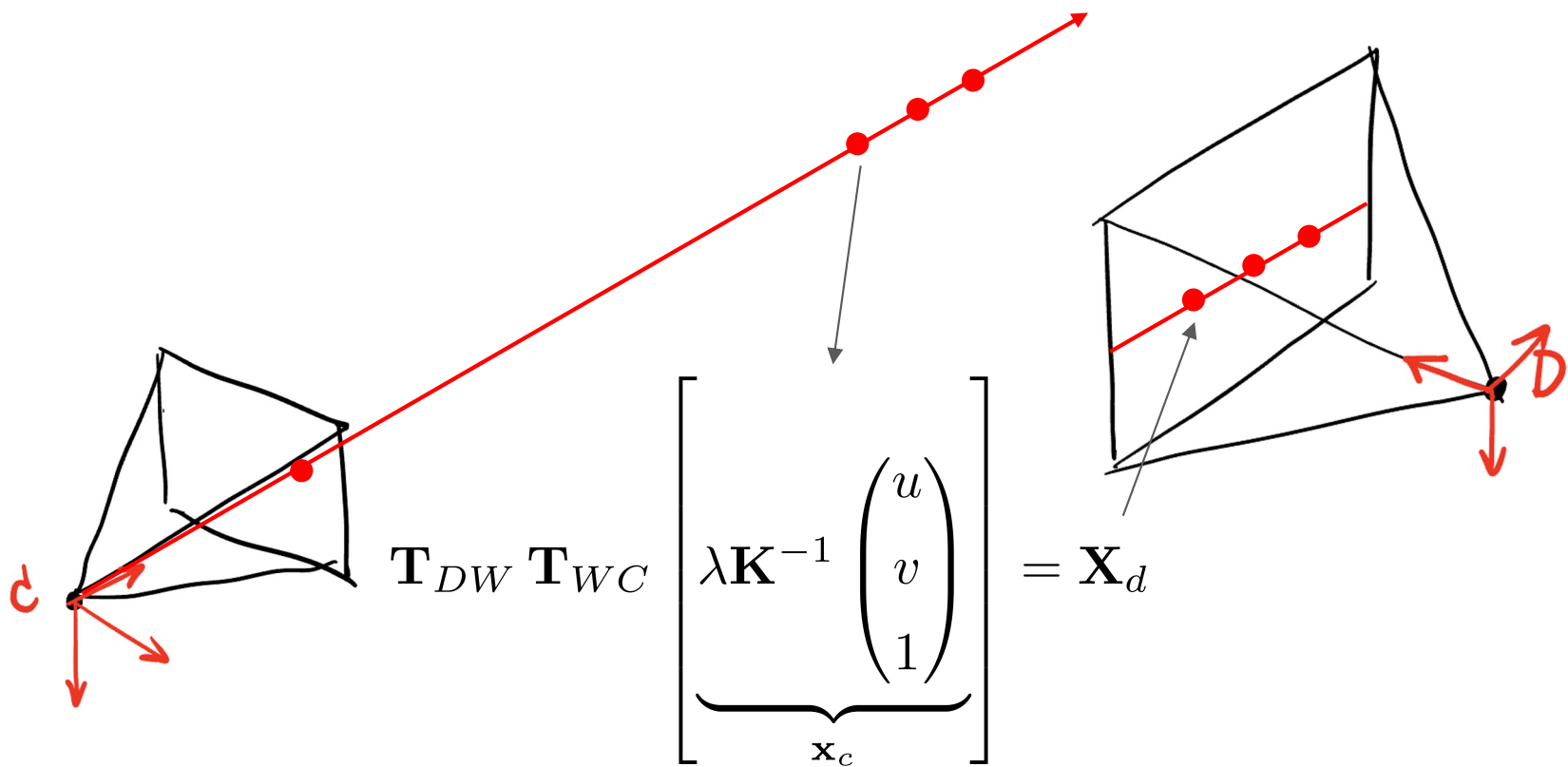
Points along the ray projected on the same pixel



From a pixel => we can recover a ray



Seeing a ray from a different camera



Let's do part 2